Enterprise RAG & Vector Database Architecture Guide

Chunking, Hybrid Search, Pinecone, ChromaDB & LlamaIndex

SkillForge AI Engineering Series - Certified Curriculum Materials

1. Document Ingestion & Recursive Chunking

Optimal chunk size: 512-1024 tokens with 10-20% chunk overlap to preserve boundary context.
Metadata enrichment: Tag chunks with document ID, creation date, author, and section headers.
Multi-modal embeddings: Support tables, code blocks, and markdown hierarchy parsing.

2. Vector Indexing & Similarity Search

Cosine Similarity measures angle regardless of magnitude; ideal for normalized text vectors.
HNSW Hierarchical Navigable Small World provides sub-5ms approximate nearest neighbor query.
Hybrid Search: Combine BM25 keyword search with dense vector embeddings for maximum recall.

3. Re-Ranking & Generation Synthesis

Re-Ranking: Use cross-encoder models e.g., Cohere ReRank, BGE-Reranker to filter top-k chunks.
Context Window Packing: Inject top-5 reranked passages into prompt context for grounded generation.